

자율주행미들웨어응용

Autonomous Driving Middleware Applications

Jaerock Kwon, PhD

Associate Professor

Electrical and Computer Engineering

University of Michigan-Dearborn

Week 1

Middleware, drive-by-wire, and the ROS 2
computation graph

This course, in one slide

- You learn ROS 2 by **finishing a real vehicle**
- MRider: a 12 V ride-on car becoming a drive-by-wire autonomous platform
- The simulator already drives, maps, and navigates
- The vehicle **does not exist yet** — parts are being ordered now

Not exercises. Every deliverable closes a named gap in a real project.

What does not exist today

Missing	Status in the repository
<code>firmware/</code>	No firmware at all
<code>mitt_hardware</code>	The plugin that makes sim and real the same stack
Measured geometry	"NOTHING IN THIS FILE HAS BEEN MEASURED"
Real tests	29 tests exist. All 29 are linters

By December these exist because **you** built them.

The semester

Weeks		
0	Last week	ROS fundamentals — orientation
1–3	Foundations	Individual labs, everyone the same
4	No class — 개천절	Self-study + Lab 4
5	Description to motion	Lab 5
6–10	Four tracks	Electrical · Chassis · Simulation · Software
11–14	Merge ladder	Subsystems combine, then full integration
15	Final demo	12/21

1. What middleware is for

A robot is not one program

- Camera driver · LiDAR driver · localiser · planner · controller · safety monitor
- Different authors, different languages, different rates, sometimes different machines

ROS 2 gives you:

- Named, typed channels instead of ad-hoc sockets
- Discovery — processes find each other unaided
- Introspection — list, inspect, record, replay at runtime
- Python and C++ in one graph

What it costs you

Coupling that used to be a **compile error** is now a **runtime condition**.

- Topic names must agree
- Message types must agree
- QoS profiles must agree
- Domain IDs must agree

When they disagree, mostly **nothing happens, and nothing complains**.

2. Drive-by-wire

Replacing the driver

Mechanical steering and throttle become **electrically commanded** equivalents.

- Steering → gearmotor + absolute angle sensor closing a position loop
- Throttle → motor driver taking a signed command

Three consequences follow, and they shape everything.

Three consequences

1. Actuation is physical, and has limits

$\pm 22.5^\circ$ of steering. A rate limit. A stall current. Software cannot wish these away.

2. Feedback is measured, not known

Every number the autonomy sees came from a sensor with noise, latency, and blind spots.

3. A human must always be able to take it back

Not a feature — an architectural requirement outranking everything else.

The command path

```
joystick → /cmd_vel_joy ─┐  
                           │→ twist_mux → ackermann_steering_controller → wheels  
navigation → /cmd_vel ─┘ (arbitrates)      (kinematics)
```

`twist_mux` priorities: **joystick 100**, navigation 10.

A human at the sticks wins.

The feedback path



- One encoder, on one motor shaft, of two motors wired in parallel
- Cannot see: gearbox backlash · tyre slip · the other wheel

Which is why it gets fused rather than trusted. (Week 6)



twist_mux is not the safety system.

And the design documents say so out loud.

Real authority is electrical

Layer	Works if the firmware has crashed?
Hardware E-stop — cuts traction	Yes
Relay MUX — reverts to stock steering	Yes
Hardware RC signal MUX	Yes
SBUS override into the controller	No

A software mux dies with the software it was meant to protect you from.

The principle

A safety mechanism that shares a failure mode
with the thing it protects against
is not a safety mechanism.

Software presenting itself as a safety system, then failing silently,
is worse than nothing — because people trust it.

3. Rates are contracts

Numbers you will keep meeting

Value	Meaning
<code>update_rate: 100</code>	Controller manager, so it is never the bottleneck
<code>reference_timeout: 0.5</code>	A command older than 0.5 s is not a command
≥ 50 Hz	Required DBW status rate from the vehicle
≤ 100 ms p95	Joystick-to-wheel latency budget

A system that meets its functional spec and misses its timing spec is **broken**.

On a vehicle, it is dangerous.

4. How ROS 2 fails

It fails by going quiet

Today you will open a terminal **without** sourcing `setup_env.sh`, and observe:

- **7 topics instead of 21**, and one node instead of the whole graph
- **Exit code 0**
- **Zero bytes on stderr**
- Nothing in any log

The simulator is running perfectly the whole time.

The habit to build now

*When something is **invisible** rather than **broken**, suspect the environment before you suspect your code.*

- A crash hands you a stack trace and a line number
- This hands you a **plausible wrong answer**: most of the graph is there, so it looks like one node crashed — and you go hunting for it

Weeks 3 and 4 add two more members to this family.

Lab 1 — Bring Up the Twin

1. Prove your environment — `check_env.sh` must report `FAIL: 0`
2. Launch the simulator; confirm both controllers `active`
3. Map the computation graph — `ros2 node list`, `topic info`, `rqt_graph`
4. Make it move — publish to `/cmd_vel_joy`
5. **Break it on purpose** — the unsourced terminal, and **partial** blindness

Deliverables: env check · graph dump · topic info · `rqt_graph.png` · written answers · AI declaration

Due before Week 2 (9/21)

Before you leave

- **Source `setup_env.sh` in every terminal** — it isolates your machine to loopback.
It takes **two** variables: Gazebo ignores the ROS one entirely
- Read M1 — Intro to MRider & ROS 2
- If setup is still broken, **talk to me now**, not next week

Next week: nodes, topics, packages — and you write the first node this repository has ever had.